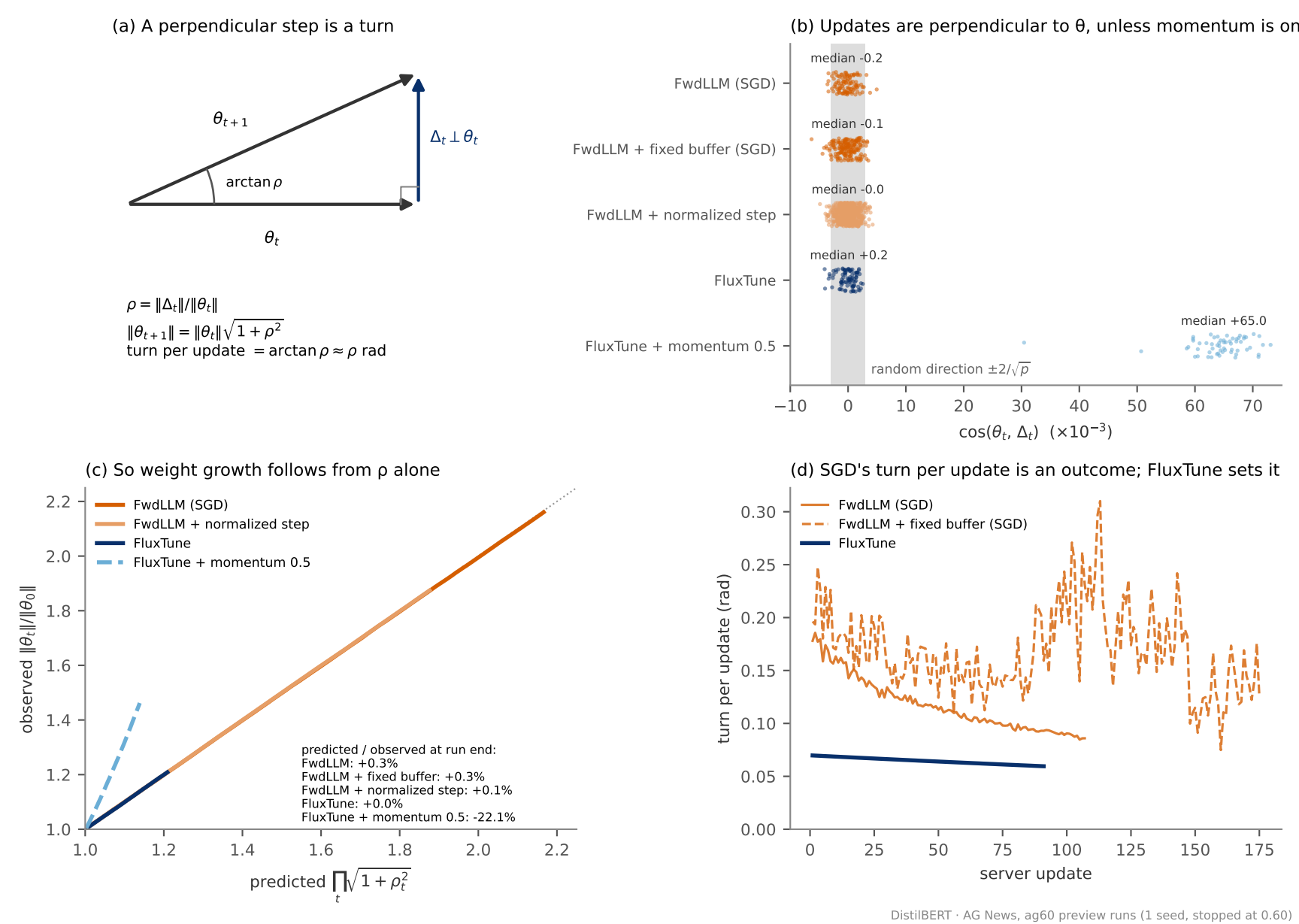

(a) A perpendicular step is a turn
θ_{t+1}
$\Delta_t \perp \theta_t$
$\arctan \rho$
θ_t
$\rho = \|\Delta_t\| / \|\theta_t\|$
$\|\theta_{t+1}\| = \|\theta_t\| \sqrt{1+\rho^2}$
turn per update $= \arctan \rho \approx \rho$ rad
(b) Updates are perpendicular to θ, unless momentum is on
FwdLLM (SGD)
FwdLLM + fixed buffer (SGD)
FwdLLM + normalized step
FluxTune
FluxTune + momentum 0.5
median -0.2
median -0.1
median -0.0
median +0.2
median +65.0
random direction $\pm 2/\sqrt{p}$
−10 0 10 20 30 40 50 60 70
cos(θ_t, Δ_t) (×10^−3)
(c) So weight growth follows from ρ alone
FwdLLM (SGD)
FwdLLM + normalized step
FluxTune
FluxTune + momentum 0.5
observed $\|\theta_t\| / \|\theta_0\|$
predicted / observed at run end:
FwdLLM: +0.3%
FwdLLM + fixed buffer: +0.3%
FwdLLM + normalized step: +0.1%
FluxTune: +0.0%
FluxTune + momentum 0.5: -22.1%
1.0 1.2 1.4 1.6 1.8 2.0 2.2
predicted $\prod_t \sqrt{1+\rho_t^2}$
(d) SGD's turn per update is an outcome; FluxTune sets it
FwdLLM (SGD)
FwdLLM + fixed buffer (SGD)
FluxTune
turn per update (rad)
0.00 0.05 0.10 0.15 0.20 0.25 0.30
0 25 50 75 100 125 150 175
server update
DistilBERT · AG News, ag60 preview runs (1 seed, stopped at 0.60)